ASSIGNMENT NO 1

1. **Print Hello World**

Code:

```
#include<iostream>
using namespace std;
int main(){
cout<<"Hello World!!"<<endl;
return 0;
}
```

o/p:-

```
pgcp-ac@dac:~/Desktop/cpp_demos2$ ./hello.out
Hello World!!
pgcp-ac@dac:~/Desktop/cpp_demos2$
```

2. **Add two numbers/binary numbers/characters**

Code: (Two Numbers)

```
#include<iostream>
using namespace std;

int main(){
int num1, num2, result;
cout<<"Enter 2 numbers: ";  //text for input of 2 nos
cin>>num1>>num2;            //input of 2 nos
result=num1+num2;           //addition of nos (right to left)
cout<<"Addition is: "<<result<<endl;        //displaying result
return 0;
}
```

o/p:-

```
pgcp-ac@dac:~/Desktop/cpp_demos2$ ./addnum.out
Enter 2 numbers: 10
20
Addition is: 30
pgcp-ac@dac:~/Desktop/cpp_demos2$
```

Code: (Two Binary Numbers)

```
//Binary Addition
#include<iostream>
using namespace std;
int addBinary(int x, int y){
while(y!=0){
int carry = x & y;        //AND operater gives carry
x = x ^ y;                //XOR operator gives sum without carry
y = carry<<1;             //shift carry
}
return x;
}

int main(){
int num1,num2, res;
cout<<"Enter 2 nos: "<<endl;
cin>>num1>>num2;
res = addBinary(num1,num2);
cout<<"Binary Adddition is: "<<res<<endl;
return 0;
}
```

o/p:-

```
pgcp-ac@dac:~/Desktop/cpp_demos2$ ./binaryaddition.out
Enter 2 nos:
5
7
Binary Adddition is: 12
pgcp-ac@dac:~/Desktop/cpp_demos2$
```

3. **Calculate compound interest**

Code:

```
#include<iostream>
#include<cmath>
using namespace std;

double compundinterest(double P, double R, double N)
{
double amount;
amount = P * pow((1+(R/100)),N);
amount = amount - P;
return amount;
}

int main(){
double P, R, N;
double CI;
cout<<"Enter Principal amount: "<<endl;
cin>>P;
cout<<"Enter Rate of Interest: "<<endl;
cin>>R;
cout<<"Enter Number of Years: "<<endl;
cin>>N;
CI = compundinterest(P, R, N);
cout<<"Compound Interest: "<<CI<<endl;
cout<<"Total Amount: "<<CI+P<<endl;
return 0;
}
```

o/p:-

```
pgcp-ac@dac:~/Desktop/cpp_demos2$ ./compoundinterest.out
Enter Principal amount:
10000
Enter Rate of Interest:
10
Enter Number of Years:
2
Compound Interest: 2100
Total Amount: 12100
pgcp-ac@dac:~/Desktop/cpp_demos2$
```

4. **Calculate power of a number**

Code:

```
#include<iostream>
using namespace std;

int power(int base, int expo){
int res=1;
for(int i=0; i<expo; i++)
{
res *= base;
}
return res;
}

int main(){
int base, expo, result;
cout<<"Enter a base: ";
cin>>base;
cout<<"Enter exponent";
cin>>expo;
result=power(base, expo);
cout<<"Power of "<<base<<" ^ "<<expo<<" is: "<<result<<endl;
return 0;
}
```

o/p:-

```
pgcp-ac@dac:~/Desktop/cpp_demos2$ ./power.out
Enter a base: 2
Enter exponent6
Power of 2 ^ 6 is: 64
pgcp-ac@dac:~/Desktop/cpp_demos2$
```

// Function Assignments

5. **Swap two numbers**

Code:

```
#include<iostream>
using namespace std;

void swapnos(int& x, int& y){          //using reference variable
x = x + y;
y = x - y;
x = x - y;
}
int main(){
int A, B;
cout<<"Enter 2 nos: "<<endl;
cin>>A>>B;
cout<<"Before Swap A: "<<A<<" B: "<<B<<endl;
swapnos(A, B);
cout<<"After Swap A: "<<A<<" B: "<<B<<endl;
return 0;
}
```

o/p:-

```
pgcp-ac@dac:~/Desktop/cpp_demos2$ ./swapnos.out
Enter 2 nos:
12
40
Before Swap A: 12 B: 40
After Swap A: 40 B: 12
pgcp-ac@dac:~/Desktop/cpp_demos2$
```

6. **Calculate area of rectangle**

Code:

```
#include<iostream>
using namespace std;

int areaRect(int len, int brd){
return len*brd;
}

int main(){
int l, b;
cout<<"Enter length and breadth in cm: "<<endl;
cin>>l>>b;
cout<<"Area of rectangle is: "<<areaRect(l, b)<<"cm^2"<<endl;
return 0;
}
```

o/p:-

```
pgcp-ac@dac:~/Desktop/cpp_demos2$ ./arearect.out
Enter length and breadth in cm:
20
30
Area of rectangle is: 600cm^2
pgcp-ac@dac:~/Desktop/cpp_demos2$
```

7. **Get 4 digit number from user, write a function to calculate sum of digits of the 4-digit number:**

a. Using loop

Code:

```
#include<iostream>
using namespace std;

int sumofdigits(int no)
{
int rem;
int sum=0;
while(no>0)
{
rem = no%10;
```

```
no = no/10;
sum+=rem;
}
return sum;
}

int main(){
int num;
int sum;
cout<<"Enter a 4-digit number: "<<endl;
cin>>num;
sum = sumofdigits(num);
cout<<"Sum of Digits of number: "<<sum<<endl;
return 0;
}
```

o/p:-

```
pgcp-ac@dac:~/Desktop/cpp_demos2$ ./sumofdigit.out
Enter a 4-digit number:
1234
Sum of Digits of number: 10
pgcp-ac@dac:~/Desktop/cpp_demos2$
```

b. Using recursion

Code:

```
#include<iostream>
using namespace std;

int sumofdigits(int num){
if(num==0)
{
return 0;
}
return (num%10)+sumofdigits(num/10);
}

int main(){
```

```
int no;
cout<<"Enter a 4 digit number: "<<endl;
cin>>no;
cout<<"Sum of digits of: "<<no<<" is "<<sumofdigits(no)<<endl;
return 0;
}
```

o/p:-

```
pgcp-ac@dac:~/Desktop/cpp_demos2$ ./sumofdigitRecursion.out
Enter a 4 digit number:
1234
Sum of digits of: 1234 is 10pgcp-ac@dac:~/Desktop/cpp_demos2$
```

8. **Write a recursive function to obtain n numbers of a Fibonacci series 0 1 1 2 3 5 8 13 21 34 55 89...**

Code:

```
#include<iostream>
using namespace std;

int fibonacci(int num){
if(num==0)
{
```

```
return 0;
}
else if(num==1)
{
return 1;
}
else{
return fibonacci(num-1) + fibonacci(num-2);
}
}

int main(){
int n;
cout<<"Enter a number: "<<endl;
cin>>n;
cout<<"Fibonacci Series is: "<<endl;
for(int i=0; i<n; i++)
{
cout<<fibonacci(i)<<"\t";
}
return 0;
}
```

o/p:-

```
pgcp-ac@dac:~/Desktop/cpp_demos2$ ./fibonacci.out
Enter a number:
11
Fibonacci Series is:
0       1       1       2       3       5       8       13      21      34      55
```

9. **Write a function to find the binary equivalent of this number using recursion.**

Code:

```
#include<iostream>
using namespace std;

void binary(int no){
if(no==0)
{
```

```
return;
}
binary(no/2);
cout<<no%2;
}

int main(){
int num;
cout<<"Enter a decimal number: "<<endl;
cin>>num;
if(num==0)
{
cout<<"0"<<endl;
}
binary(num);
return 0;
}
```

o/p:-

```
pgcp-ac@dac:~/Desktop/cpp_demos2$ ./binaryEquivalent.out
Enter a decimal number:
7
111pgcp-ac@dac:~/Desktop/cpp_demos2$
```